

Attention Manipulation for Hallucination Mitigation

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ABSTRACT

Object hallucination remains a critical challenge for large vision-language models (LVLMs), undermining their applicability in safety-critical domains. We present an attention manipulation framework, AtMAN-in-Box, that extends AtMan by incorporating spatial grounding through bounding box-based attention modulation. Our approach significantly reduces hallucination in both single- and multi-object scenarios. Experimental results show a consistent performance gain across precision, recall, and hallucination suppression metrics. Our method not only enhances visual grounding but also improves the interpretability and robustness of LVLMs.

INTRODUCTION

While LVLMs demonstrate impressive performance in tasks such as image captioning and visual QA, they often suffer from hallucination—describing objects not present in the image. This work builds on AtMan, a saliency-based method for interpreting LVLM predictions, and proposes spatially-aware attention modulation to improve grounding and mitigate hallucinations.

METHODOLOGY

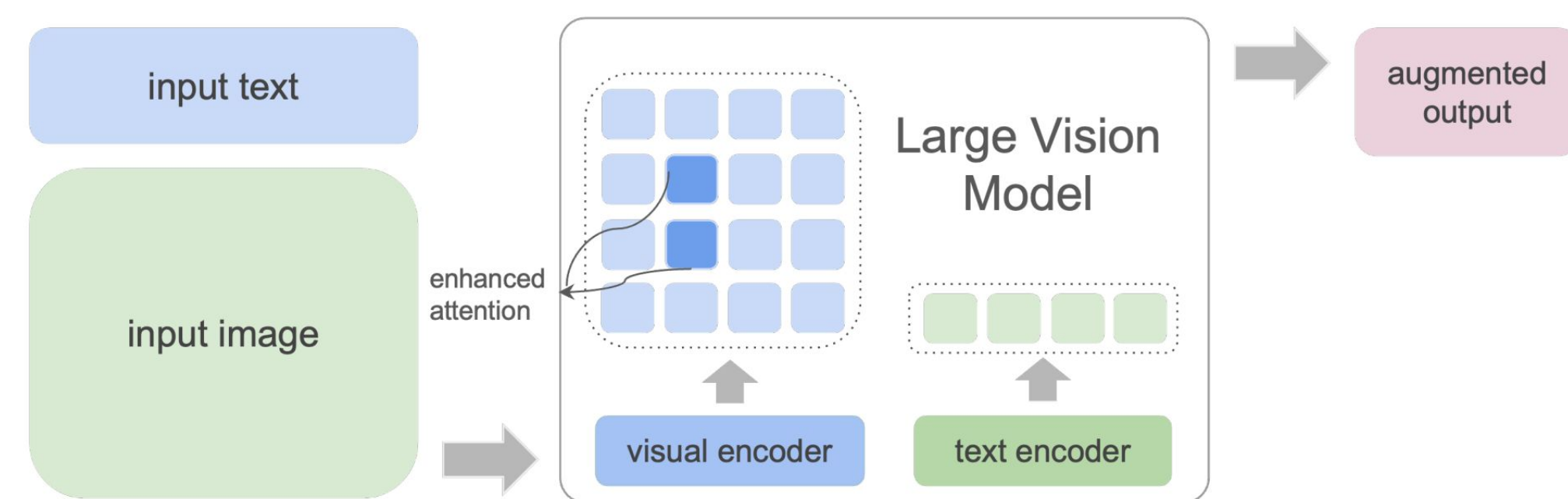


Figure 1: Pipeline of the proposed method.

AtMAN: Attention Manipulation

We replicate AtMan, which manipulates attention weights to visualize important tokens for model predictions. Attention is suppressed or amplified at specific positions and the resulting change in loss (influence score) is used to generate saliency maps.

YOLO Integration

To improve spatial grounding, we detect objects using YOLOv8 and incorporate bounding boxes to guide attention manipulation. This approach forms the core of our AtMAN-in-Box extension.

Multi-Object Handling

For complex scenarios, we introduce a dataset-specific bounding box processing pipeline. We standardize box color, re-label detected objects as `obj{i}`, and cap the number of boxes to 5 per image. The processed annotations are used to guide multi-object attention.

New Dataset

We construct a new dataset curated for hallucination evaluation, hosted on Hugging Face. It consists of real-world images with five labeled hallucination instances in each image.



Figure 3: Example labeled image from our dataset.

RESULTS

Single-Object Evaluation

We compare the baseline AtMan implementation with and without YOLO-enhanced bounding box grounding:

	Without YOLO	With YOLO
mAP	68.58	69.82
mAR	48.68	52.39
IQ-mAP	68.72	69.63
IQ-mAR	50.28	52.90

YOLO integration improves both precision and recall.

Qualitative Improvements

Bounding boxes from YOLO correct several ground truth errors in OpenImages. We visualize this improvement through saliency maps and bounding box overlays.

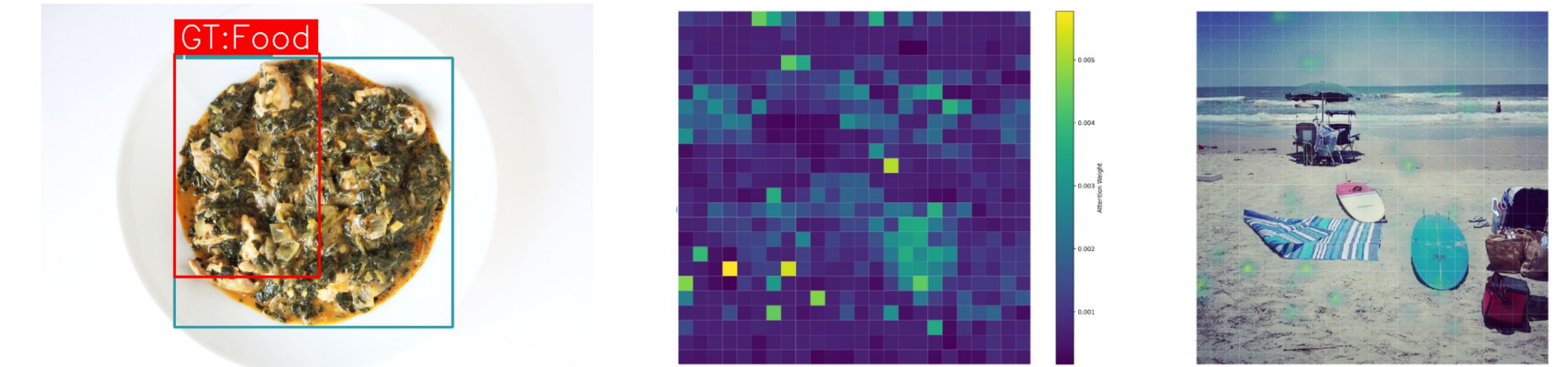


Figure 2: YOLO generates better bounding boxes (left).

Figure 3: Visualization of AtMAN method as a saliency map (right).

Multi-Object Evaluation

We measure hallucination reduction across various object settings, using the MOH benchmark:

Model	Seen Wild	Seen Hom.	Seen Het.	Unseen Wild	Unseen Hom.	Unseen Het.
LLaVA-1.5	21.26%	52.40%	7.69%	13.96%	31.88%	3.98%
+AtMAN-in-Box	23.80%	55.60%	8.50%	16.00%	35.60%	4.50%

New Dataset Performance

We evaluated hallucination classification accuracy using our new dataset, and got an accuracy of 41.43%.

RECOMMENDATIONS

- Incorporate spatial priors using detection outputs for improved visual grounding.
- Use saliency-based attention modulation for model interpretability and debugging.
- Benchmark hallucination using diverse scenarios (single-object, multi-object, seen/unseen).

CONCLUSION

We introduce AtMAN-in-Box, a spatial-aware attention manipulation framework that integrates YOLO object detection to enhance grounding and reduce hallucinations in LVLMs. Our method shows measurable improvement in both quantitative metrics and qualitative grounding, particularly in multi-object settings.